

Employment

Graduate Research Assistant	NC State University	April 2025 – Present
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- Deployed containerized genomic platform on GCP Cloud Run (Docker, Cloud Build CI) with JBrowse 2 genome browser; enabled 100+ researchers to visualize 500GB+ datasets by eliminating on-premise infrastructure.
- Improved ML model accuracy by 12% via grid-search hyperparameter sweeps and sliding-window feature engineering on multivariate biological sensor time-series.

Product Engineer Intern	Isomer AI	May 2025 – May 2026
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- Shipped 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST APIs; 923 commits via GitHub Actions CI/CD across 7 deployments.
- Owned PostgreSQL, Redis, Auth0 RBAC, Terraform, Docker, and Sentry stack; triaged and resolved production incidents by tracing failures across async microservices.

Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
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- Improved document processing accuracy by 25% by building OCR + NLP classification pipelines that parsed unstructured health claim documents into structured fields for downstream automation.
- Automated AWS and GCP releases via GitHub Actions CI/CD; configured Grafana dashboards for observability and led red-teaming to surface model reliability gaps pre-production.

AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
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- Reduced ticket resolution time by 15% by training an NLP multi-class classifier that auto-routed support tickets to the correct team based on predicted category and urgency.
- Cut ticket-processing latency by 20% by replacing synchronous DB polling with Kafka async ingestion feeding Spark micro-batch jobs across partitioned Hadoop HDFS.

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
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- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.

Mumbai, India	Mumbai University	Aug 2020 – May 2024
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- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.

Projects

KubeServe	Kubernetes LLM Inference Platform	2026
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- Built shared LLM inference platform on Kubernetes: KEDA autoscaler on queue-depth metrics, Kopf CRD operator routing requests across Claude, GPT-4o, and Gemini, circuit breaker + EWMA load balancing across replicas.
- Instrumented OTel distributed tracing, Prometheus, and Grafana to optimize p95 latency under 500-VU k6 load; shipped via Helm + ArgoCD GitOps with 88 pytest tests enforced in GitHub Actions CI.

VectorLift	Semantic Search & Ranking Engine	2026
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- Built Kafka-driven corpus ingestion pipeline into Elasticsearch and Qdrant; served semantic search via FastAPI REST API with per-stage latency tracking across 6 configurable pipeline configurations.
- Improved retrieval quality by fine-tuning bi-encoder (MS MARCO) and cross-encoder re-ranker (MiniLM); evaluated with NDCG@10, MRR@10, and paired bootstrap significance testing.

Languages and Technologies

- **Languages:** Python, TypeScript, Java, JavaScript, SQL, Bash, C
- **AI & LLMs:** Claude API, GPT-4o, Gemini, prompt engineering, Temporal.io agentic workflows, Kafka
- **Backend & APIs:** FastAPI, NestJS, Spring Boot, REST APIs, event-driven architecture, Elasticsearch, Qdrant
- **Infrastructure:** AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, PostgreSQL, Redis, MongoDB, Grafana

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog